Laser Map Explorer (LaME) Workshop

Duration: Half day (09:00–13:00)
Build: `LaME v0.9.0-workshop`
Datasets: Three garnet maps, one pyrite map

> Note
>
> Fill in your own date, venue and contact details above before printing.

What today covers 1
Before you start 2
How to read this handout 3
The datasets 3
1. Opening the project 4
2. Choosing fields 4
3. One field at a time 6
4. Two fields, then three 7
5. Regions and filters 8
6. Clustering 9
7. Mineral formulae across three garnets 11
8. Pyrite — your turn 12
9. Notes and the methods report 13
10. Take it home 14
Known limitations in this build 14
If you fall behind 15
Appendix A: Profiles 15
Appendix B: Getting your own data in 16
Getting help 16

What today covers

LaME turns LA-ICP-MS element maps into something you can rapidly explore: filter, plot, cluster, calculate, and get mineral formulae out the other end.

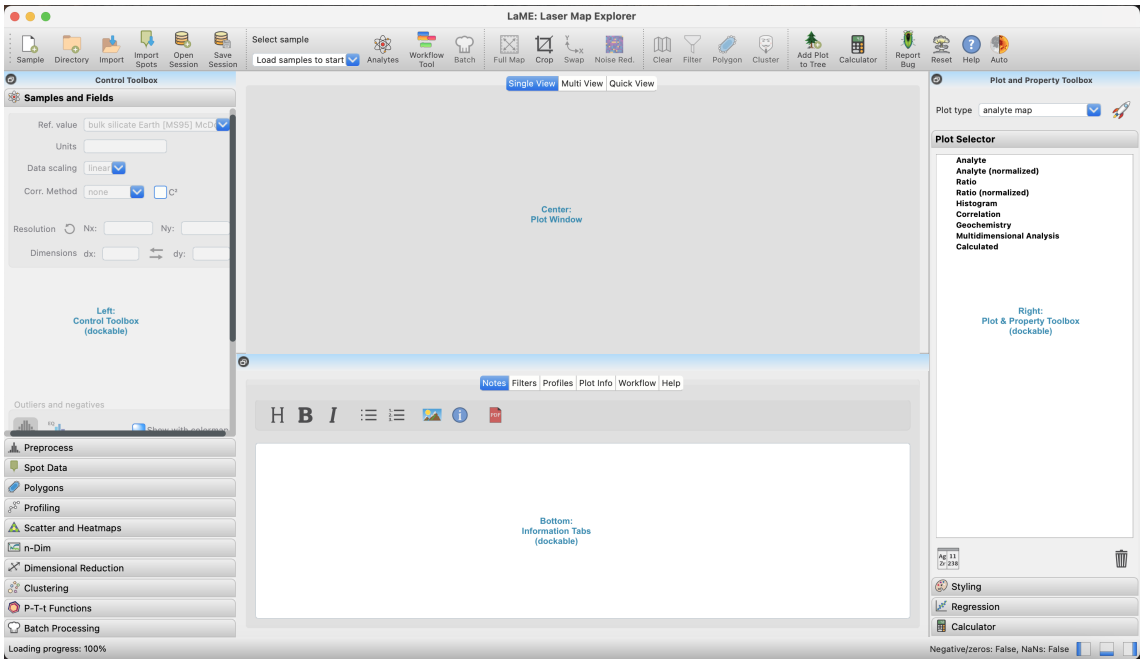
We work through two threads.

Garnet — the quantitative thread. One sample to learn on, then the same analysis across three garnets, ending with mineral formulae and a map of the end-member distribution.

Pyrite — the qualitative thread. A completely different mineral system, where you drive. Find the pyrite, cluster it, and ask whether arsenic tracks gold.

Core – XRF. Use many of the same tools to investigate XRF intensity maps. Turn semi-quantitative data into something that you can interpret rapidly and with confidence.

LaME is beta software and this handout is honest about that. Where something is unfinished you will see a **Not in this build** box telling you what you would see and what to do instead. Those boxes are not apologies — they are there so you do not waste the workshop discovering them yourself. A consolidated list is in Known limitations in this build.



The LaME main window immediately after opening. There are four main areas in the window: (1) Control Toolbox, left; (2) plot canvas, center; (3) Styling Toolbox and Plot Tree, right; (4) bottom tabs; (5) the toolbar.

Before you start

You need two things, both of which should already be on your machine:

1. **LaME installed** — the `.dmg` (macOS) or `.zip` (Windows) from the release link you were emailed.
2. **The workshop folder**, unzipped **to your Desktop with its structure intact**:

```
LaME-workshop/
■■■ data/
■   ■■■ RM02.lame.csv          garnet, 78,228 px
■   ■■■ WOS-02.lame.csv        garnet, 123,255 px
■   ■■■ TR1-07.lame.csv        garnet, 59,916 px
■   ■■■ 72C-1.lame.csv         pyrite, 61,617 px
■■■ 00-open.lame_project.json      + 00-open/
■■■ 02-plots.lame_project.json     + 02-plots/
■■■ 04-clusters.lame_project.json  + 04-clusters/
■■■ 06-pyrite.lame_project.json    + 06-pyrite/
■■■ handout.pdf
```

> **Warning**
>
> Do not move files out of that folder or rename the directories beside each `.lame_project.json`. Project files find their data by **relative** path, so the structure is what makes them work on your machine rather than only on the presenter's.

How to read this handout

Numbered steps are things you do. Do them in order — a few of them only work in one order, and those are called out.

> Checkpoint
>
> **Grey boxes like this are things you should see.** If your screen does not match, do not debug it: open the project file named in the box and carry on. You will lose nothing that later sections need.

> Not in this build
>
> **Boxes like this are known limitations.** They tell you what you would see if you tried, and what to do instead.

Try this paragraphs at the end of a section are optional. They exist so that if you finish early you have somewhere to go that does not put you ahead of the group.

The datasets

Three garnets and a pyrite, all real data.

Sample	Pixels	Mineral	Notes
`RM02`	78,228	Garnet	Garnet–chlorite schist, Walter–Outalpa Shear Zone, South Australia. 28 analytes: full REE suite plus Ca, Fe, Mg, Mn, Al, Ti, Cr, P, Li, Zr, Th, U, Pb.
`WOS-02`	123,255	Garnet	33 analytes. Adds **Si**, Na, Sr, Ba and a second Pb isotope.
`TR1-07`	59,916	Garnet	33 analytes, same suite as WOS-02. The smallest and quickest of the three.
`72C-1`	61,617	Pyrite	31 analytes: S, Fe and the chalcophile suite — Co, Ni, Cu, Zn, **As**, Se, Ag, Cd, Sn, Sb, Te, **Au**, Hg, Tl, Pb, Bi.

What these datasets **cannot** do — worth knowing up front, because all three are properties of the data rather than of the software:

- **RM02 has no silicon.** Garnet is $X_3Y_2Z_3O_{12}$, and Si normally fills the tetrahedral Z site. Without it, LaME fills Z with aluminium instead. We use this deliberately in 7. Mineral formulae across three garnets — it is the clearest demonstration you will get of what the stoichiometry calculation needs from your data.
- **None of them can do Lu–Hf dating.** That needs $^{176}Lu/^{177}Hf$ and $^{176}Hf/^{177}Hf$. These maps have ^{175}Lu and at most one Hf isotope. The capability is real — you will see a result from a different sample — but no amount of clicking will produce an isochron from today's data.
- **The pyrite map is uncalibrated.** Values are counts per second, not ppm. Everything we do with it — maps, filters, clusters, biplots — is valid on CPS. Mineral formulae and chondrite normalisation are **not**, so do not run the Stoichiometry dock on it.

One practical wrinkle: the three garnets were run on different setups, so **isotope names differ between samples**. RM02 has `Fe57`, `Ti49`, `Cr53`; WOS-02 and TR1-07 have `Fe56`, `Ti47`, `Cr52`. When a step names a field, check the sample you are actually on.

1. Opening the project

Goal: get a map on screen in under two minutes, with everyone seeing the same thing.

1. Launch LaME. First launch takes a little while — it is unpacking, not hung.
2. **Home** toolbar page → **Open Project**.
3. Navigate to `Desktop/LaME-workshop/` and choose `00-open.lame_project.json`.

That is all. You did not open the CSV directly, and you will not need to today — a project file carries the sample, the selected fields and the window state together, which is why it makes a reliable starting point and a reliable *recovery* point.

Checkpoint 0

The title bar reads **LaME - 00-open**. The Control Toolbox on the left is on its **Preprocess** page. There is a field map of RM02 in the centre and a small histogram of the same field beside it.

Not matching? You are at the first step; just try the open again.

Not in this build

Look at the **Plot** box at the top of the Control Toolbox. It reads `field map`. The entry directly below it, `gradient map`, produces nothing in this build — either a warning about noise reduction, or silence and the previous plot left on screen.

We now leave this page and do not come back. Everything else today happens on pages whose plot lists contain no unfinished entries.

1. In the Control Toolbox, switch from **Preprocess** to **Field Viewer**.

What you just learned. Projects, not CSVs, are how you save and resume work in LaME. The Control Toolbox page you are on determines which plot types are offered.

2. Choosing fields

Goal: control which analytes are in play, build a ratio, and calculate a new field.

Nothing in LaME plots an analyte you have not selected. That selection also drives what the correlation matrix shows and what clustering uses, so it is the first real control you have.

Selecting analytes and a ratio

1. Click **Analytes** in the pinned toolbar row. The *Select Analytes and Ratios* dialog opens on a square grid of your analytes.

2. The **diagonal** cells are the analytes themselves. Click along it to select `Mn55`, `Y89`, `Sm147`, `Eu153`, `Gd157`, `Ca43`, `Fe57`, `Mg24`. Selected cells highlight.
3. The **off-diagonal** cells are ratios. Above the diagonal gives row ÷ column; below gives column ÷ row. Find the cell where `Y89` meets `Mn55` and click it once.
4. Your choices appear in the **Selected** list on the right, with columns *Field*, *Scaling* and *Normalized*. The ratio appears as `Y89 / Mn55`.
5. Click **Done**.

The background colours in that grid are the correlation matrix — so the dialog is already telling you which analytes co-vary before you plot anything.

Setting the reference

1. Back in the Control Toolbox, find the **Ref.** row and set it to a chondrite — `CI chondrite [AG89]`.

Do this **before** the next part. Normalised fields divide by this reference, and the default is bulk silicate Earth, which is not what you want for a REE anomaly.

Calculating Eu*

Europium anomalies need a field that is not in your data: the interpolated Eu you would expect from its neighbours. That is what the Calculator is for.

1. **Analysis** toolbar page → **Calculator**.
2. In the saved-formula box, choose `Eu*`. The formula appears:

   ```
   {Analyte.Eu153_N}/({Analyte.Sm147_N}*{Analyte.Gd157_N})^(1/2)
   ```

   Fields are written `{FieldType.Field}`. The `_N` suffix means reference-normalised — which is why the reference had to be set first.
3. Click **Calculate**.
4. Plot it: Control Toolbox **Field Viewer**, plot type `field map`, then set the colour field type to `Calculated` and the field to `Eu*`.

Not in this build

The Calculator's keypad offers `case()` and `otw()` buttons for conditional expressions. Conditionals do not work in this build — you will get *"could not evaluate conditional statement"*.

The shipped Calculator documentation also describes a `case(...)` **function**. That syntax has never existed in any version. **Ignore the ``case()`` section of the Calculator docs.** Arithmetic, logs, powers and roots all work fine.

Checkpoint 1

The Plot Tree on the right has a **Calculated** branch containing `Eu*`. Your canvas shows a smooth, low-contrast Eu* map — much flatter than a raw element map, because you have divided out the overall REE level.

Not matching? Open `02-plots.lame_project.json` — it has the fields, the ratio and Eu* already built.

1. **Save Project** in the pinned toolbar. Save it next to the workshop folder under your own name.

From here on, start each section by saving. You are building your own recovery points alongside the ones we supply.

What you just learned. Field selection gates everything downstream. Ratios come from the analyte grid; anything more complex comes from the Calculator. Normalised fields depend on the reference material being set first.

3. One field at a time

Goal: read a single element's distribution properly before comparing elements.

Histograms

1. Control Toolbox **Field Viewer**, plot type `histogram`.
2. Set the **X** field to `Mn55`.
3. In the **Histogram** group, set *Histogram type* to `PDF` and tick **Show KDE**.
4. Try `log-scaling`. Trace-element distributions are usually closer to log-normal than normal, and the shape change tells you whether yours is.
5. Set the colour field type to `Cluster` or `ROI` later in the day and this same histogram splits by group — a quick way to see whether two zones really differ.

Correlations

1. Plot type `correlation`.
2. Set **Method** to `Pearson`, then compare against `Spearman`. Pearson assumes a linear relationship; Spearman only assumes monotonic, so a large gap between them is itself informative.
3. Tick **C²** to show correlation squared — the shared-variance view.

The matrix covers exactly the analytes you selected in section 2. Change the selection and the matrix changes with it.

> Not in this build
>
> The correlation matrix **ignores filters**. Later today you will filter the map; the correlation plot will not respond. The same applies to the ternary map.
>
> To see a filter's effect, use `field map`, `histogram` or `roi map`.

What you just learned. Histogram shape tells you which scaling to use. Correlation method choice is a real decision, not a default. Both read from your field selection.

4. Two fields, then three

Goal: move from distributions to relationships, and back onto the map.

1. Control Toolbox → **Scatter and Heatmaps**. The axis rows now have labels: **X**, **Y**, **Z** and **Color**.

Biplot

1. Plot type `scatter`. Set **X** to `Mn55` and **Y** to `Y89`. Leave **Z** empty.
2. Set **Color** to `Ca43`. A third variable as colour is usually more informative than a third axis.

Heatmap

1. Plot type `heatmap`, same X and Y. With 78,000 points, overplotting hides the structure; binning reveals it.
2. In the **Biplot and Ternary** group, try the *Heatmaps* statistic — `counts`, then `median`. `counts` shows where the data is; `median` shows what a third variable does there.

Ternary

There is no separate "ternary" plot type. A ternary appears when you fill in Z.

1. Plot type `scatter`, then set **Z** to `Fe57`. X, Y and Z must be three different fields.
2. With `X = Mn55`, `Y = Ca43`, `Z = Fe57` you have a compositional ternary of the three major garnet cations.

Ternary map

1. Plot type `ternary map`, same three fields.
2. In the **Ternary Map** group, either pick a **Colormap** or set the four colour swatches — one per vertex plus one for the centre.
3. Click **Ternary map**.

Each pixel is now coloured by where it sits in the ternary. This is the single most useful plot in LaME for seeing compositional zoning, because it puts three elements on one map without you choosing which matters most.

Checkpoint 2

Your Plot Tree contains five plots: a field map, a histogram, a correlation, a scatter and a ternary map. The canvas shows the ternary map with clear zoning — the garnet core should be a visibly different colour from its rim.

Not matching? Open `02-plots.lame_project.json`.

Try this: double-click any leaf in the Plot Tree. It restores that plot — type, fields and styling together. The tree is a record of your session, not just a list.

What you just learned. Scatter and heatmap answer the same question at different densities. Adding Z turns either into a ternary. The ternary map is how you get a three-element relationship back onto the spatial map.

5. Regions and filters

Goal: restrict the analysis to the material you actually care about.

This is the part of LaME with the most moving parts, so read step 1 before clicking.

Filters live inside regions of interest. An ROI is a named, saved definition; a filter is one rule inside it. You select the ROI first, then add filters to it. If you add a filter with no ROI defined, LaME quietly creates one for you — which is convenient once and confusing the second time, when it starts asking which region you meant.

Your first ROI and filter

1. **Processing** toolbar page → **Filters**. The Masking Toolbox opens at the bottom on its **Filters** tab.
2. In the **ROI** table at the bottom of that tab, click **Add ROI**. A new row appears. Double-click its *Name* and call it `garnet`. **Leave this row selected.**
3. Now in **Filter Settings** on the left: set *Field type* to `Analyte` and *Field* to `Mn55`. The Min and Max boxes fill with that field's range.
4. Narrow it. Either type values, or use the percentile spin boxes beside them — the percentile route is usually faster for a first pass.
5. Set *Operator* to `and`.
6. Click **Add filter**.

The filter appears as a row in the table on the right, with a **Use** checkbox, and the map updates.

1. Add a second filter the same way on a different element — `Ca43`, say. Because the `garnet` row is still selected, it joins the same region.

Seeing what you removed

1. Control Toolbox **Field Viewer** → plot type `roi map`.

Excluded pixels are drawn grey. This is the plot to use whenever you want to check what a filter actually did — it shows you the mask itself rather than its effect on a value.

Polygons

Filters cut by composition. Polygons cut by location.

1. Masking Toolbox → **Polygons** tab.
2. Flip the toggle switch at the left of its toolbar to enter polygon mode. The plot type collapses to `field map` only while you are drawing.
3. Click **Create a new polygon**, then click round a feature on the map to place vertices.
4. The *In/out* column on each row controls whether the polygon keeps or removes what is inside it.
5. Use **Move a control point**, **Add a vertex** and **Remove a vertex** to adjust.

> Not in this build
>
> Polygons are deliberately **excluded from clustering**. Drawing a polygon around a grain and re-clustering will not restrict the clustering to that grain.
>
> To scope an analysis spatially, use an ROI. That is what the clustering and stoichiometry sections do.

> Not in this build
>
> Two toolbar buttons to avoid: **Crop** and **Full Map**. Cropping is incomplete in this build, and the *Full Map* button that would undo it raises an error instead. Nothing today needs either; if you crop by accident, reopen your saved project.
>
> **Save filter** is also unavailable in the packaged build — it tries to write inside the application bundle. Your filters are saved in your *project*, which is what matters.

> Checkpoint 3
>
> The ROI table has one row named `garnet` with a non-zero **% Filtered**. The `roi map` shows your garnet in colour with a grey excluded rim.
>
> *Not matching?* Open `04-clusters.lame_project.json`, which has the ROI already built — it also has clusters, which you are about to need anyway.

What you just learned. ROI first, then filters into it. `roi map` is how you verify a mask. Polygons are spatial, filters are compositional, and only filters and ROIs scope downstream analysis.

6. Clustering

Goal: let the data propose its own zones, then test whether they are real.

1. Control Toolbox → **Clustering**. Opening the page runs clustering immediately with the current settings and plots the result, so you will see groups before you have chosen anything.

k-means

1. Set **Method** to `k-means` and **No. clusters** to `4`.
2. Plot type `cluster map`.
3. Click **Randomize** beside the **Seed** box and watch the map. k-means starts from random centres; if your zones survive several seeds they are robust, and if they reshuffle they were not there.

> **Not in this build**
>
> The **Distance** combobox on this page has no effect — the distance-metric code is not active in this build. Changing it recomputes and gives an identical result.
>
> Also: `HDBSCAN` works as a method, but combining it with the `cluster performance` plot type silently does nothing. HDBSCAN discovers its own cluster count, so there is no *k* to sweep. Use `cluster performance` with k-means or fuzzy c-means only.

Changing what clustering sees

Clustering uses exactly the fields you selected in section 2 — so changing that selection changes the clusters.

1. Click **Analytes** and add two or three REE you had left out — `La139`, `Ce140`, `Nd146`.
2. Click **Done**. Because you are on the Clustering page, it recomputes and replots automatically.
3. Compare against what you had. Adding light REE will usually sharpen a growth-zoning boundary, because those elements partition strongly during garnet growth.

> **Note**
>
> That automatic recompute only happens while you are on the **Clustering** or **Dimensional Reduction** page. From anywhere else, change the selection and then press **Update Plot**.

Clustering on principal components

With 11 correlated analytes, much of what k-means sees is the same signal repeated. Reducing first often gives cleaner groups.

1. Control Toolbox → **Dimensional Reduction**. Opening the page runs PCA.
2. Set **Method** to `PCA: Principal component analysis`. Look at the `variance` plot to see how many components you actually need — usually three or four carry nearly everything.
3. Go back to **Clustering**. Tick the **PCA** checkbox and set **No. basis** to the number you just decided on.
4. Re-cluster.

Run PCA *before* ticking that box. If no PCA scores exist, LaME falls back to the raw analytes without telling you, and the comparison you were trying to make silently does not happen.

Naming and reusing clusters

1. Masking Toolbox → **Clusters** tab. The table lists your clusters with *Name*, *Link*, *Color*, *% Total* and *% Filtered*.

2. Rename a cluster in place — `core`, `rim`, `inclusion`. These names follow through into the legend, the Notes report and the stoichiometry scope list.
3. Recolour one by clicking its *Color* cell.
4. Tick two clusters that you think are the same zone and click **Link Clusters** — they then behave as one class.
5. Tick the cluster you want to work on and click **Create Region**. It becomes an ROI, which is how you will scope the next section.

> Checkpoint 4
>
> A `cluster map` with four k-means groups, the **PCA** box ticked, and a Clusters table with four named rows. At least one cluster has been turned into an ROI, so the ROI table in the Filters tab now has a second row.
>
> *Not matching?* Open `04-clusters.lame_project.json`. Its clusters restore from disk without recomputing, so it loads quickly.

What you just learned. Clustering reads your field selection, so curating that selection is the main lever you have. PCA first usually improves it. Clusters become ROIs, and ROIs are how you scope everything downstream.

7. Mineral formulae across three garnets

Goal: get apfu, site occupancies and end-member proportions, and map them.

LaME can convert composition into a mineral formula — site by site, with iron split between Fe^{2}■ and Fe^{3}■ by charge balance. The chemistry behind this is the most heavily validated part of LaME, cross-checked against published mineral-recalculation workbooks for garnet, feldspar, olivine and pyroxene.

> Important
>
> **Scope it to one region before you touch anything else.** The calculation runs pixel by pixel, on the interface thread, and it re-runs every time you change any setting. Across a whole 78,000-pixel map that is minutes of frozen window, several times over. Scoped to one cluster it is seconds.

RM02 — and what happens without silicon

1. **Analysis** toolbar page → **Stoichiometry**. The *Stoichiometric Calculator* opens.
2. **First**, in the **Compute scope** group: set *Filter* to the ROI you made from a cluster, and tick **only that region** in the list below. New regions start unticked deliberately — leave the others alone.
3. Now the **Settings** group: *Mineral* = `Garnet`, *Input basis* = `ppm`, *Redox method* = `droop_1987`, *BDL treatment* = `zero`.
4. Leave **Compare all redox methods** switched **off**. It triples the work.
5. Click **Run**.

New fields appear under field type `Stoichiometry`:

Field	Meaning
`Grt_Z`, `Grt_Y`, `Grt_X`	Cations per formula unit on each site — tetrahedral, octahedral, dodecahedral
`Grt_pyrope` … `Grt_uvarovite`	End-member proportions: pyrope, almandine, spessartine, grossular, andradite, uvarovite
`Grt_dominant`	Which end member dominates at each pixel — a categorical field

1. Plot it: `field map`, colour field type `Stoichiometry`, field `Grt_dominant`. It draws with a labelled swatch legend rather than a colour bar, because the values are categories rather than amounts.

Now look at `Grt_Z`. It should be close to 3.0, filled by silicon. **On RM02 it is not** — RM02 has no Si, so LaME has filled the tetrahedral site with aluminium instead, following the fallback order in the garnet configuration. The end-member percentages that come out are correspondingly wrong.

This is worth seeing rather than being told. The stoichiometry calculation is only as good as the elements you measured, and a missing major element does not produce an error — it produces a plausible-looking answer that is incorrect.

The same garnet, with silicon

1. Switch sample using the selector in the pinned toolbar: choose `TR1-07`.
2. Make a quick ROI on it — one filter on `Mn55` is enough; you do not need to re-cluster.
3. Scope the Stoichiometry dock to that ROI and **Run** again.
4. Compare `Grt_Z` with what RM02 gave you. With `Si29` present it should sit near 3.0, and the end-member proportions are now meaningful.
5. Repeat on `WOS-02` if time allows. It is the largest of the three, so keep the region small.

Checkpoint 5

At least two garnets have `Grt_*` fields, and you have a `Grt_dominant` map with a swatch legend on screen. `Grt_Z` is near 3.0 for TR1-07 and clearly not for RM02.

Not in this build

If the **Mineral** dropdown is empty and outlined in red, the application cannot find its mineral configuration files. Tell the presenter — there is a workaround.

Try this: the **Region summary** table at the bottom of the dock gives per-region statistics. Switch *Region source* between `ROI` and a cluster column to compare formulae between zones you identified in section 6.

What you just learned. Scope first, always. Site totals are your sanity check on whether the calculation had the elements it needed. `Grt_dominant` maps mineral classification back onto the map.

8. Pyrite — your turn

Goal: apply what you have done to a different mineral system, with less guidance.

This section has fewer numbered steps on purpose. You have the tools now; the question is new.

The question. 72C-1 is a pyrite-bearing sample. Orogenic gold systems often carry gold in arsenian pyrite, so arsenic and gold should track each other — *if* they are in the pyrite rather than in something else. Find out.

Remember this map is **uncalibrated**: counts per second, not ppm. Relative patterns are fine; absolute values and mineral formulae are not.

1. Open `06-pyrite.lame_project.json`, or add `data/72C-1.lame.csv` yourself.

   You will see a field called `Time [Sec]` among the analytes. It is an artefact of the export. Ignore it.

2. **Find the pyrite.** Field maps of `Fe57` and `S34`. Pyrite is FeS■, so it should stand out sharply against the silicate matrix in both.

3. **Isolate it.** Add an ROI named `pyrite`, then filters on `Fe57` and `S34` to keep only the high-Fe, high-S pixels. Check your work with `roi map` — grey is what you excluded.

4. **Cluster within it.** Go to the Clustering page and cluster with 3 or 4 groups. You are now asking whether the pyrite is chemically uniform or zoned.

5. **Test the question.** Scatter and Heatmaps page, plot type `scatter`: **X** = `As75`, **Y** = `Au197`, **Color** = your cluster field.

   If As and Au are coupled, you get a trend. If one cluster sits off that trend, that population has a different gold-hosting relationship from the rest — which is a result worth having.

6. **Three elements at once.** Set **Z** to `Co59` or `Ni60` for a ternary, then switch to `ternary map`. Co and Ni substitute for Fe in pyrite, so a ternary map often shows growth zoning that no single element reveals.

Checkpoint 6

A `pyrite` ROI isolating the sulfide, clusters computed inside it, and an As–Au biplot coloured by cluster.

Try this: try the same biplot *without* the ROI, on the whole map. The silicate matrix points usually swamp the pyrite trend completely — a concrete demonstration of why the filtering step was not optional.

What you just learned. The same four tools — ROI, filter, cluster, biplot — carry across mineral systems unchanged. Uncalibrated data is perfectly usable for pattern work. Filtering before plotting is often what makes a relationship visible at all.

9. Notes and the methods report

Goal: leave with a record of what you did, not just the plots.

1. **Log** toolbar page → **Notes**. An editor window opens — titled *reST Editor* — holding a notes file tied to this sample inside your project folder.

2. Click **Record** in the same toolbar page. The status-bar *Notes* light turns red.

3. Now make a plot — anything. It is captured into the notes automatically as a figure.

4. In the Notes toolbar, open the **Insert** page and use **Formatted Info**. It offers:

    - Sample info
    - List analytes used
    - Current plot details
    - Filter table
    - PCA results
    - Cluster results
    - Diffusion results
    - Stoichiometry results

    Insert **List analytes used**, **Filter table** and **Cluster results**. Each writes a formatted block pulled from your live session — the actual analytes, the actual filters, the actual clusters.

That is the methods paragraph of a paper, assembled from what you really did rather than from what you remember doing a month later.

> Not in this build
>
> There is also a **Capture** toggle, and a Workflow designer with a drag-and-drop block canvas. Capture currently records only sample changes as reusable blocks — so treat it as a **methods report generator**, which it does well, rather than as a way to record your clicks into a re-runnable workflow. Hand-built workflows in the designer do run; recording them does not yet.

What you just learned. Notes are per-sample and live inside the project. Record captures figures as you go. Formatted Info blocks pull real state, so the record stays honest.

10. Take it home

1. **Save Project**.
2. Close the project — **File** → **Close Project**.
3. Reopen it. Your fields, ratios, calculated fields, ROIs, filters, polygons, clusters and notes all come back.

That round trip is the thing to trust: the project file is the unit of work, and it survives being moved between machines because it stores paths relative to itself.

Known limitations in this build

Everything from the boxes above, in one place.

Feature	What you will see	What to do instead
`gradient map` plot type	A warning about noise reduction, or nothing at all	Not available; use `field map`
Calculator conditionals	*"could not evaluate conditional statement"*	Arithmetic only. **Ignore the ``case()`` section of the Calculator docs** — that syntax never existed
Crop / **Full Map**	Crop is incomplete; *Full Map* raises an error instead of undoing it	Avoid both. Reopen your project if you crop by accident

`profile` and `polygon` plot types	Blank canvas, no error	Drive profiles from the **Profiles** dock (Appendix A)
Correlation and ternary map	Both ignore filters and masks entirely	Check filter effects on `field map`, `histogram` or `roi map`
Adding a filter with no ROI selected	First one silently creates an ROI; the second asks which region you meant	Create and select the ROI row first
Polygons + clustering	Clustering ignores polygons by design	Scope with an ROI instead
Cluster **Distance** combobox	No effect on the result	Leave it alone
HDBSCAN + `cluster performance`	Silently does nothing	Use k-means or fuzzy c-means for that plot
Save filter preset	*"could not save filter table"* in the packaged build	Your filters are saved in the project, which is what matters
Stoichiometry on a whole map	Minutes of frozen window	Scope to one ROI or cluster first
Lu–Hf dating	Needs ^{1}■■Lu/^{1}■■Hf and ^{1}■■Hf/^{1}■■Hf, absent from today's data	Implemented, but not demonstrable on these samples
Workflow **Capture**	Records only sample changes as blocks	Use it as a methods-report generator
Thermometry / Barometry tabs	Empty method dropdowns	Not implemented

If you fall behind

Open the nearest project file and carry on. You lose nothing that later sections need.

File	Picks up at
`00-open.lame_project.json`	Section 1 — RM02 loaded, nothing else done
`02-plots.lame_project.json`	Section 4 — fields, ratio and Eu* built
`04-clusters.lame_project.json`	Section 7 — ROI and clusters ready
`06-pyrite.lame_project.json`	Section 8 — pyrite sample loaded

File → **Open Project**, pick the file, carry on. Do not spend the workshop debugging.

Appendix A: Profiles

Profiles extract a transect across a map — the standard way to show garnet zoning as a function of radius. It works, but the route to it is not the obvious one.

> Not in this build
>
> Do **not** select `profile` from the plot-type list. It produces a blank canvas. Profiles are driven entirely from the Profiles dock.

1. Get a `field map` on screen.
2. **Analysis** toolbar page → **Profiles**.
3. Flip the toggle switch at the left of the dock's toolbar. Plot type collapses to `field map` while profiling is active.
4. Click **Create new profile and set control points**, then click along the map — for a garnet, rim through core to the far rim.
5. Set **Point radius** (how much to average around each point), **Interp. distance** (spacing along the transect) and **Point type** (`median + IQR` is the robust choice).
6. In **Axis Options**, set **No. subplots**, then for each subplot pick a field and click the add button. Several elements stacked against distance is how zoning is normally presented.
7. Use **Edit mode** to click out any point sitting on a crack or inclusion.
8. Save with the export button. Profiles are stored per sample in your project.

Appendix B: Getting your own data in

Today's samples were already converted to LaME's `.lame.csv` format. For your own data, the Import tool handles raw exports.

1. **Home** toolbar page → **Import Files**.
2. Choose a **Data type** and **Method** — `LA-ICP-MS` with `quadrupole`, `TOF` or `SF`, or `XRF` with `image`.
3. Point it at your directory. The metadata table fills with one row per sample.
4. Set spot size, sweep time and stage speed; LaME derives pixel spacing from them.
5. Use the per-sample **Select files** button to check which files map to which analytes, then **Import**.

> Not in this build
>
> Only `LA-ICP-MS` and `XRF` → `image` import in this build. The other data types populate the metadata table and then stop with a *"Not implemented"* message, so the table filling in is not a sign that it will work.

If a conversion gives you trouble, send the presenter a raw export and you will get a `.lame.csv` back.

Getting help

In the application. Help menu → *User Guide* or *Tutorials*. There is also a context-sensitive mode: toggle **Help** in the toolbar, then click any part of the interface to jump to its documentation.

Video walkthroughs.

- Introduction to LaME — interface, navigation, view modes
- Loading Data and Selecting Samples — import formats and analyte selection

- Visualization Fundamentals — correlation, spatial patterns
- Working with Notes

Reporting a problem. Help menu → *Report Bug*. Please include what you were doing, what you expected, and what happened. Everything in Known limitations in this build is already known — anything else is worth telling us about, and today is exactly the sort of session that surfaces new ones.